Multi-Sensory Recommendation System for Home Furniture Using Mulstimodal Deep Learning and Cross-Layer Attention

Mohan Gaddam
Data science
University of North Texas

Mohit Pavan Raj Meda
Data science
University of North Texas

Deepak Gogikar
Data Science
University of North Texas

Malavika Gudala
Data Science
University of North Texas

***Abstract*—This project introduces a multimodal recommendation system for furniture items using multi-modal deep learning integrated with cross-layer attention mechanisms to model different human senses like visual aesthetic feelings, auditory preferences, and even tactile sensations. The power of most traditional recommendation systems remains restricted to mono-modal data, such as texts or images, which impairs the power in capturing users' preferences for specific domains like interior design and e-commerce. This system hence aligns and prioritizes the huge variety of sensory inputs for framing recommendations that meet holistic sensory expectations from users. The proposed architecture is based on cross-layer attention that can process and integrate data dynamically in a multimodal way and understand the completeness of user preference. Datasets used would be from IKEA, Walmart, and Amazon, including public datasets of texture and audio data, for developing such a scalable prototype. The performance will be considered effective concerning different metrics, including precision and recall. This research aims to reshape recommendation systems and contribute significantly to cross-modal personalization while increasing user activity related to multisensory environments.**

***Index Terms*—Multi-Sensory Recommendation System, Multi-modal Deep Learning, E-commerce,cross-modal personalization**

I. Introduction

E-commerce has transformed the retail sector, giving customers unparalleled access to items across diverse sectors, including fashion, technology, and home design. Recommendation systems have become essential to the digital purchasing experience, facilitating customers' discovery of things that correspond with their interests and requirements. By recommending pertinent items, these systems not only augment user happiness but also stimulate engagement and revenue for e-commerce platforms (Ailyn, 2024). Conventional recommendation systems depend on collaborative filtering, content-based filtering, or a hybrid approach, evaluating data including user ratings, purchase history, and product characteristics. Nonetheless, as consumer expectations advance, there is an increasing desire for more immersive and personalised experiences that traditional, single-modal recommendation methods cannot provide.

A. Limitations of Single-Modal Recommendation Systems

In sectors like home goods, fashion, and interior design, customer choices are often influenced by diverse sensory experiences, including visual aesthetics, tactile sensations, and acoustic atmosphere. Notwithstanding this, most recommendation algorithms in e-commerce concentrate on single-modal data, often depending on either text or graphics. This method neglects the complex nature of human preference development, particularly in areas where sensory interactions are crucial to the consumer experience. Aesthetically pleasing sofas may disappoint consumers if the fabric's texture does not align with their comfort preferences or if it does not provide the right atmosphere in their living environment (Ailyn, 2024). Single-modal recommendation systems are inherently constrained in their ability to provide a comprehensive user experience. Systems that concentrate only on visual data are incapable of integrating additional sensory information, such as tactile qualities or auditory characteristics, which might be essential in influencing purchase choices. Text-based recommendation algorithms are equally limited, since language may be subjective and may not sufficiently represent the many sensory characteristics of a product. As a result, these systems fail to address the intricate sensory requirements of consumers, especially in areas such as home furnishings, where material, atmosphere, and aesthetic attributes have equal significance (Ailyn, 2024).

B. The Need for a Multi-Sensory Recommendation Approach

To mitigate these constraints, there is an increasing focus on creating multi-sensory recommendation systems that use diverse sensory modalities, such as visual, aural, and tactile information. Multi-sensory systems seek to provide a more comprehensive and authentic depiction of items via the analysis of several types of input data. In the realm of home furnishings, a multi-sensory recommendation system would evaluate not just the visual aesthetics of an item but also the tactile qualities of its materials and the acoustic environment it generates. Consequently, such a system would provide a more thorough comprehension of user preferences, resulting in suggestions that are more congruent with customers' real-life product experiences (Bao et al., 2024). Utilising breakthroughs in multi-modal deep learning and cross-layer attention processes, multi-sensory recommendation systems may surpass the limitations of conventional systems. By integrating many sensory inputs, these systems transition from a static, single-modal framework

to a dynamic, user-centric model that reflects the complex nature of human preferences. Multi-sensory recommendation systems signify a substantial advancement in the development of personalised e-commerce experiences, enabling suggestions that connect more profoundly with consumers (Bao et al., 2024).

II. Problem Statement

This dissertation tackles the task of developing a recommendation system that integrates multi-sensory inputs to improve personalisation in the home furnishings industry. The project seeks to create a recommendation system that incorporates three data types: visual (picture aesthetics), auditory (ambient sound preferences), and tactile (texture and material sensations). The system employs deep learning models with cross-layer attention methods to analyse and align various inputs, therefore delivering more accurate and pleasing suggestions for users. The system employs a multi-sensory approach to surpass the constraints of single-modal recommendation systems, therefore providing a more enriched and immersive online purchasing experience (Chen et al., 2021).

A. Research Aim and Objectives

This project aims to address the deficiency of existing recommendation systems in capturing the multi-sensory factors that affect user preferences in categories such as home furnishings. Current methods are unable to include non-visual sensory cues, such as texture and atmosphere, which are essential for developing a comprehensive recommendation experience. This technological gap results in suggestions that may seem visually attractive but ultimately do not fulfil the user's sensory expectations, leading to poor user experiences (Chen et al., 2021).

1) Objectives:

- Construct a multi-sensory recommendation system that amalgamates visual, aural, and tactile data to provide tailored product recommendations in the home furnishings sector.
- Utilise multimodal deep learning methods and cross-layer attention processes to efficiently align and process various sensory inputs (Fang et al., 2024).
- Improve user happiness and suggestion precision by developing a recommendation system that embodies the comprehensive sensory preferences of users.
- Assess the efficacy of multi-sensory data integration in enhancing the relevance and personalisation of product suggestions in contrast to single-modal systems (Fang et al., 2024).

This dissertation advances the evolution of next-generation recommendation systems that can comprehend and cater to users' multi-sensory preferences. The suggested system in this research seeks to augment suggestion precision while also elevating user pleasure via a more immersive and contextually relevant purchasing experience. This paper examines the technological problems of incorporating multi-sensory data into a deep learning framework, providing insights into the advantages and constraints of this methodology in e-commerce (Galmarini & Visalli, 2024).

This dissertation proposes a multi-sensory recommendation system that incorporates visual, aural, and tactile inputs to more accurately represent consumers' comprehensive preferences in the home furnishings sector. This project seeks to enhance recommendation algorithms by transcending single-modal frameworks, therefore fostering a more immersive and personalised e-commerce experience for customers (Galmarini & Visalli, 2024).

III. Literature Review

The concept of multimodal learning pertains to the amalgamation and analysis of several data kinds, or modalities, including text, pictures, audio, and sensory information such as touch. Multimodal learning in recommendation systems has garnered considerable interest for its ability to capture a more thorough and nuanced knowledge of user preferences, especially in sectors reliant on sensory-rich experiences, such as fashion, interior design, and entertainment. In contrast to single-modal systems that depend on restricted input types (e.g., text-only or image-only data), multimodal systems integrate many data sources, enabling a more profound comprehension of users' intricate and diverse preferences (Galmarini & Visalli, 2024).

Multimodal data integrates inputs that each provide distinct information on user preferences. A multimodal recommendation system in fashion may evaluate both visual aesthetics (from photographs) and written product descriptions to provide tailored suggestions. Multimodal learning may address deficiencies in single-modal systems that may neglect essential aspects of intricate user preferences. This method improves the precision and pertinence of recommendations, becoming them more responsive to the user's comprehensive requirements. Ultimately, multimodal learning augments the efficacy of recommendation systems, providing a more immersive and tailored user experience via the integration and alignment of many information layers from varied data sources (Galmarini & Visalli, 2024).

A. Cross-Layer Attention Mechanisms

Sreevidya et al. (2024) presents a novel methodology for multimodal data fusion that facilitates clustering in applications like personality and affective computing. This inter- and intra-modal data fusion methodology, using a binary search algorithm, establishes a central clustering layer that integrates attention processes. The resultant fused signal enables the identification of task-related elements that are distinctive to personality prediction but are not adequately recognised or comprehended by the relevant experts. An outstanding feature of Sreevidya's et al. (2024) work is her comprehensive utilisation of diverse data sources, encompassing audio, visual, and textual cues, which are analysed using advanced deep learning techniques, specifically convolutional neural networks (CNNs) and long short-term memory (LSTM) models. However, while

the study outlines a comprehensive framework for personality-aware computers, it may encounter scaling issues when implemented with bigger datasets or diverse practical contexts. The substantial computational expense associated with deep networks and attention techniques is an additional drawback. According to the authors of the article, transparency on challenges faced in developing scalable interoperable models is anticipated to mitigate such shortcomings. Furthermore, there are no explicit examples or validation for applying this strategy outside the emotional paradigm and into other multimodal domains.

F. Zhang et al. (2023) research, "Cross-Layer Connected Network with Adaptive Attention Mechanism for 3D Multi-Organ and Tumour Segmentations from CT," tackles a significant challenge in medical image processing, namely in the segmentation of tiny tumours from 3D CT images. The research creatively combines cross-layer connections and adaptive attention to address class imbalance, specifically enhancing accuracy for tiny tumours. The integration of Anti-Aliased Pooling (AAP) and Region of Interest (ROI) adaptive attention techniques signifies substantial progress, enabling enhanced feature extraction and spatial recovery. Nonetheless, despite its accomplishments in segmentation precision, F. Zhang's et al. (2023) model may face constraints in generalisability over varied datasets beyond liver and brain tumours, since variances in medical images might impede adaptation. Moreover, the model's intricacy, characterised by cross-layer linkages and adaptive attention processes, may hinder its implementation in clinical environments with constrained computing resources.

B. Applications of Multimodal Deep Learning in Recommendation Systems

Gordon and Greece (2024) paper, Application of Multimodal Deep Learning in Sentiment Analysis for Recommendation Systems, introduces a multimodal deep learning model designed to improve recommendation systems via the integration of text, picture, and user behavioural data. This novel methodology emphasises sentiment analysis to enhance recommendation precision and user contentment. A major aspect of Gordon and Greece's (2024) work is its empirical validation across many public datasets, showcasing its superiority over conventional sentiment analysis and recommendation techniques. The amalgamation of multimodal data facilitates a more profound comprehension of user preferences, thus improving suggestion precision. Nonetheless, a restriction exists in the model's reliance on big labelled datasets, which may pose challenges in practical applications when multimodal sentiment data is limited or unstructured. Moreover, while this model shows potential for sentiment-oriented suggestions, it may need additional refining to accommodate other recommendation contexts outside sentiment-driven applications.

Mu and Wu's (2023) study, Multimodal Movie Recommendation System Using Deep Learning, tackles the difficulties in movie recommendation systems, particularly cold-start and sparse data problems, by using multimodal data and deep learning techniques. Mu and Wu's (2023) solution employs the MovieLens datasets to integrate multimodal analysis, revealing latent aspects of films and users, hence significantly enhancing recommendation accuracy compared to conventional collaborative filtering methods. The decrease in RMSE scores underscores the efficacy of this method in delivering tailored suggestions. Nonetheless, while Mu's approach addresses sparse data challenges to a degree, it may still encounter difficulties with scalability and generalisation across many genres or media kinds outside films. Furthermore, the model's dependence on the extensive MovieLens dataset may limit its application to other datasets or environments characterised by less organised multimodal information. This study offers significant insights into alleviating data sparsity in movie suggestions; nonetheless, it would be enhanced by further testing on diverse and real-world datasets to evaluate its wider application.

C. Role of Large Language Models (LLMs) in Recommendation Systems

The paper by Vats et al. (2024), Exploring the Impact of Large Language Models on Recommender Systems: An Extensive Review, emphasises the transformational influence of Large Language Models (LLMs) in enhancing recommendation systems via advanced language understanding and reasoning skills. Vats asserts that LLMs have distinct benefits over conventional recommenders, particularly in their capacity to comprehend sophisticated language and provide contextually relevant suggestions. This transition allows LLMs to connect deficiencies in user interaction data and provide suggestions derived from inferred preferences based on linguistic patterns. This article's significant strength lies in its thorough examination of the paradigm change brought about by LLMs, addressing factors such as domain adaptability and iterative learning. Nonetheless, the article recognises problems, including the sensitivity of LLMs to prompt modifications and the possibility of inadvertent suggestions. Although the analysis is comprehensive, it might be enhanced by a more in-depth examination of real applications across various industrial contexts, which may provide further insights into the scalability and resilience of LLM-based recommendation systems.

Bao et al. (2024) essay, Large Language Models for Recommendation: Past, Present, and Future, provides a tutorial-oriented analysis of LLMs in recommendation systems, emphasising their development and capacity to transform item retrieval and ranking processes. Bao offers a comprehensive analysis of how LLMs might improve recommender systems via features such as conversational engagement, generalised learning, and content creation. The article's strength is its examination of growing issues in the sector, including the need for reliability, efficiency, and dependable assessment measures for LLM-driven recommenders. Bao's anticipatory strategy, delineating prospective research trajectories, is a significant enhancement, since it promotes the ongoing advancement of LLMs to tackle these difficulties. The paper might benefit from a more comprehensive consideration of particular case studies or applications in which LLMs have been effectively

incorporated into recommendation systems, therefore providing practical context for readers and augmenting the material's usefulness for applied research.

D. Previous Methodologies for Multisensory Data Integration

The paper by Rawat and Rawat (2015), titled Multi-sensor Data Fusion by a Hybrid Methodology – A Comparative Study, examines the amalgamation of multi-sensor data via a hybrid methodology that integrates Rough Set (RS) theory with a Back-Propagation Neural Network (BPNN). Rawat's methodology seeks to enhance data fusion precision in intricate nonlinear systems, essential for applications in wireless sensor networks. The integration of RS with BPNN exhibits improved prediction accuracy relative to conventional BPNN models. This hybrid technique is especially beneficial since RS minimises data redundancy, hence enhancing the neural network's learning process. Nonetheless, while the research demonstrates enhanced accuracy, it is constrained by its restricted emphasis on RS and BPNN, possibly neglecting further fusion methodologies that might further augment system resilience. Furthermore, the essay might be enhanced by examining how this hybrid model may be modified for real-time applications where processing speed is essential.

Luo and Chang (2011) essay, Multisensor Fusion and Integration: A Review of Approaches and Its Applications in Mechatronics, offers an extensive examination of multi-sensor fusion methodologies and their applications in many domains, especially in mechatronics. Luo classifies fusion techniques into estimation, classification, and inference approaches, examining their individual advantages and drawbacks. This review's notable strength is in its extensive examination of MFI applications, ranging from industrial automation to robotics, underscoring the adaptability of these approaches in managing intricate environmental data. Nevertheless, while the study presents a useful categorisation of current approaches, it fails to provide a comprehensive evaluation of their performance in actual, real-world applications, which may yield more actionable insights for practitioners. The study would benefit from a more comprehensive examination of the constraints and limits associated with applying MFI in dynamic, high-stakes contexts, such as autonomous systems, where sensor dependability is paramount.

E. Case Studies on Multimodal Recommendation Systems

Wu et al. (2022) essay, A Personalised Recommendation System for Multi-Modal Transportation Systems, examines a framework for a recommendation system that integrates concepts from recommendation algorithms and transportation engineering inside multi-modal transportation. Wu's methodology employs past user data and real-time traffic conditions to recommend ideal travel options, seeking to reconcile individual comfort with overarching societal advantages such as less traffic congestion and environmental effects. The article's strength is its hierarchical behaviour structure, which efficiently handles huge datasets by extracting multi-scale characteristics without overburdening computer resources. Nonetheless, the framework's dependence on past data and traffic circumstances may limit its relevance in situations characterised by substantial fluctuations in user behaviour or unforeseen traffic disturbances. Furthermore, while the research indicates possible societal advantages, it is devoid of actual information about the system's effects on public transport utilisation or traffic mitigation in practical contexts.

Fang et al. (2024) study, Enhanced Multimodal Recommendation Systems via Reviews Integration, presents the Personalised Multi-Preference Recommender (PMPR) model, which integrates user reviews to enhance the accuracy of multimodal recommendations. In contrast to conventional models that emphasise exterior characteristics of products, PMPR identifies internal attributes—such as cost-efficiency or simplicity of maintenance—that are often referenced in reviews but neglected in other data sources. This methodology employs a heterogeneous graph learning module for reviews and a homogeneous module for multimodal data, efficiently reducing noise by analysing each review separately using attention vectors. The efficacy of Fang's model is attributed to its capacity to accommodate varied user preferences across several data dimensions, shown by its exceptional performance on three datasets. The intricacy of the PMPR model may pose difficulties for real-time applications, when processing speed and computational efficiency are paramount. Moreover, while Fang's approach demonstrates enhanced performance, additional investigation into its flexibility across various recommendation situations would aid in substantiating its wider applicability. Summary and Gaps in Existing Research The reviewed literature highlights the progress in multimodal learning, especially via cross-layer attention mechanisms, large language models (LLMs), and hybrid data fusion methods, which have markedly improved recommendation precision, personalisation, and resilience across diverse applications. These approaches have been effective in synchronising multimodal data to enhance user comprehension, presenting opportunities in domains like as medical imaging, sentiment analysis, and transportation. Nevertheless, significant deficiencies persist. There is a need for enhanced integration of multisensory data, especially in attaining seamless alignment across various sensory modalities, such as auditory and tactile inputs. Moreover, scaling concerns and substantial computing requirements pose difficulties, particularly in real-time applications and resource-constrained environments. Additional research is necessary to improve flexibility, efficiency, and generalisability across diverse and dynamic datasets.

IV. Objectives of the Study

The primary objective of this work is to create a multimodal recommendation system that incorporates visual, aural, and tactile data to improve the quality of recommendations for home furnishings. This method seeks to overcome the shortcomings of conventional recommendation systems, which often depend on single-modal data (such photos) and do not fully include the spectrum of user preferences, by integrating many sensory inputs (Gul Khan, 2023).

The primary goal is to develop a model architecture that can independently process each sensory modality. This enables the model to discern the distinctive characteristics of each input—such as hues and forms in pictures, atmosphere in music, and material textures—yielding a more refined comprehension of user preferences (Jeong et al., 2024). The second purpose is to amalgamate these sensory attributes into a cohesive recommendation system that mirrors real-world decision-making processes, whereby consumers evaluate several sensory dimensions. This multi-modal strategy aims to provide suggestions that better fit with user demands, therefore enhancing overall happiness and engagement (Jeong et al., 2024).

The ultimate goal is to assess the model's performance to gauge its efficacy and pinpoint areas for improvement. The research seeks to enhance the domain of recommendation systems, making them more immersive and tailored for consumers by accomplishing these objectives (Jeong et al., 2024).

V. Data Collection

This dissertation used three independent datasets, each corresponding to a different sensory modality (pictures, audio, and textures), to create a multi-sensory recommendation system for home furnishings. The datasets were selected to correspond with the visual, aural, and tactile elements essential for understanding user preferences in home furnishings. The following is a comprehensive overview of each dataset, including their origins, attributes, and the preparation procedures used to ready them for the model (Lai, 2024).

A. Image Dataset: Furniture Image Dataset

This project use the Furniture Image Dataset from Kaggle, which has a thorough assortment of photographs categorised into five types of furniture: Almirah, Chair, Refrigerator, Table, and Television. The collection is organised hierarchically, with a primary folder that comprises five subfolders, each corresponding to a furniture category containing 3000 photos. The dataset comprises 15,000 photos, serving as a substantial resource for training machine learning models to differentiate among various kinds of furniture (Lai, 2024).

Every picture in the dataset has a resolution of 96 dots per inch (dpi) and a bit depth of 24, guaranteeing adequate image quality for deep learning applications. Nonetheless, the dataset is unlabelled, without any associated information or annotations, rendering it entirely visual. Each picture was reduced to 224x224 pixels for preprocessing, standardising input dimensions for a convolutional neural network (CNN). Furthermore, pixel values were standardised to a range of 0 to 1, enhancing the model's convergence efficiency during training. This picture collection underpins the visual aesthetics of furniture products, an essential element in augmenting the realism and attractiveness of product suggestions (H. Li et al., 2023).

B. Audio Dataset: ESC-50 Environmental Sound Dataset

The selected audio dataset for this project is the ESC-50 Environmental Sound Dataset, which provides a diverse array of ambient noises from various environmental situations, hence enhancing its applicability for replicating different home and interior atmospheres. In contrast to datasets tailored to a certain context, ESC-50 comprises 2000 audio recordings categorised into 50 distinct groups, with each clip having a duration of 5 seconds. The categories are broadly classified into five principal themes: Animals, Natural Soundscapes & Water Sounds, Human Non-Speech Sounds, Interior/Domestic Sounds, and Exterior/Urban Noises. This variant allows the recommendation algorithm to account for various ambient audio settings that may correspond with customers' preferred home surroundings (K. Li et al., 2024).

The Interior/Domestic noises category include sounds such as door knocks, keyboard typing, washing machines, and clock ticks, typical of a household setting. Furthermore, natural noises, such as rainfall, ocean waves, and wind, provide a serene ambiance that is often sought for in indoor environments. The audio snippets were carefully collected from publically available recordings on Freesound.org, ensuring high-quality samples. During preprocessing, each audio clip was converted into Mel-frequency cepstral coefficients (MFCCs), a commonly used feature in audio research that represents the spectrum characteristics of sound. Thirteen MFCCs were extracted from each clip and averaged across time to generate a fixed-size feature vector. This preprocessing enabled the efficient amalgamation of audio input with visual and texture data, enhancing the system's ability to correlate objects with suitable ambient sounds (K. Li et al., 2024).

C. Texture Dataset: Describable Textures Dataset (DTD)

The Describable Textures Dataset (DTD), supplied by the Visual Geometry Group, presents an extensive array of textures categorised into 47 human-perception-inspired classifications, including "polka-dotted," "striped," and "lined." These descriptors provide a sophisticated method for comprehending textures, enabling robots to replicate human tactile perception in photographs. The collection has 5,640 photos, with around 120 images per category, including diverse texture types that may depict varied material finishes for furniture (Y. Li, 2024).

Each picture in the DTD measures between 300x300 and 640x640 pixels, guaranteeing a high fidelity of texture features, with a minimum of 90% of the texture aligning with its designated label. The photos are annotated with primary and joint characteristics, gathered via Amazon Mechanical Turk, so including a dimension of human perception into the dataset. During preprocessing, the texture pictures were shrunk to 224x224 pixels to meet the input specifications of the CNN model, and pixel values were normalised to a range of 0 to 1. This texture collection is essential for modelling the tactile characteristics of furniture, depicting various material sensations that affect user preferences (Y. Li, 2024). The datasets were obtained online from publicly accessible sources on Kaggle and the Visual Geometry Group's website. Each dataset was carefully preprocessed to ensure compatibility with the multi-modal deep learning model, which included resizing images, normalizing pixel values, and extracting audio

features. Collectively, these datasets provide the basis of the multi-sensory recommendation system, including visual, aural, and tactile preferences that allow more personalised and immersive suggestions for home furnishings. The system intends to use these varied datasets to provide a more sophisticated and nuanced user experience that reflects customers' real-life interactions with furniture (Luo & Chang, 2011).

Dataset	Source
Image Dataset	https://www.kaggle.com/datasets/udaysankarmukherjee/furniture-image-dataset
Audio Dataset	https://github.com/karolpiczak/ESC-50
Texture Dataset	https://www.robots.ox.ac.uk/ vgg/data/dtd/

TABLE I
DATA COLLECTION SUMMARY

VI. EXPLORATORY DATA ANALYSIS (EDA) AND HYPOTHESES FOR THE STUDY

A. Exploratory Data Analysis (EDA)

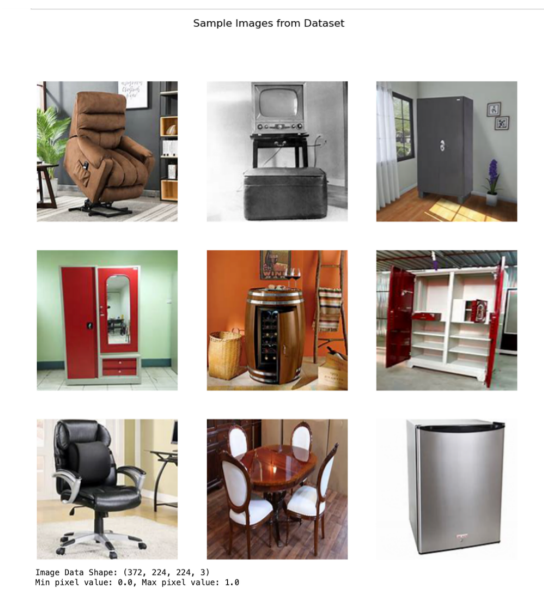

Fig. 1. EDA of Image Dataset

The picture collection comprises 372 samples across five furniture categories: Almirah, Chair, Refrigerator, Table, and Television. Every picture was downsized to a standardised dimension of (224, 224, 3) and normalised to pixel values ranging from 0 to 1, so assuring uniformity across the collection. A grid of photographs exemplifies the numerous furniture pieces, highlighting different colours and designs. This visual variety will assist the model in distinguishing between furniture kinds based on aesthetic characteristics

The audio dataset, derived from the ESC-50 collection, has 44 recordings of ambient noises. These sounds are ideal for replicating diverse ambient home environments. The waveform

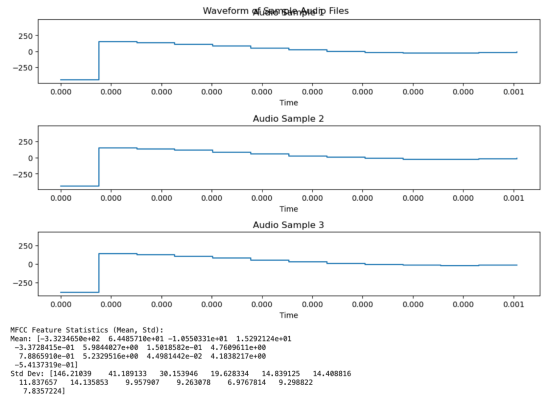

Fig. 2. EDA of Audio Dataset (in Waveform)

of each audio sample was examined to comprehend its time-domain characteristics. Moreover, Mel-frequency cepstral coefficients (MFCCs) were retrieved as features, offering spectral attributes essential for differentiating between sounds. Summary statistics, including the mean and standard deviation of MFCCs, provide an in-depth analysis of the acoustic characteristics of the dataset.

Fig. 3. EDA of Texture Dataset

The texture dataset has 110 samples classified by patterns such as "striped" and "polka-dotted." Analogous to the picture modality, each texture image was enlarged to dimensions of (224, 224, 3) and subjected to normalisation. Sample texture photos exhibit a variety of patterns that visually convey distinct tactile properties. This variation in texture aids the model in recommending goods based on material aesthetics and tactile qualities, a crucial element in furniture selection.

B. Impact of EDA on Model Architecture Design

The first results from the EDA offered significant insights into each modality, informing both the preprocessing procedures and the model architecture.

1) Image Modality: The dataset consists of 372 photos categorised into five furniture types, with each image scaled to a uniform dimension of '(224, 224, 3)'. The consistency in picture dimensions facilitates the efficient processing of inputs inside the convolutional neural network (CNN) architecture. The variety in furniture types and colours indicated the need for a model adept in extracting complex characteristics, hence a multi-layer CNN with progressively increasing filters was selected to efficiently capture these subtleties. Normalising pixel values to a range of 0 to 1 facilitated expedited and more consistent model convergence (Martínez-García & Hernández-Lemus, 2022).

2) Audio Modality: The audio dataset included 44 samples, each converted into a 13-dimensional feature vector using Mel-frequency cepstral coefficients (MFCCs). The limited dataset size and concise feature representation guided the choice to use a straightforward dense layer architecture for the model's audio branch. The MFCCs offered spectrum insights that enable the model to differentiate between varieties of ambient sound, a valuable attribute for aligning audio preferences with furniture designs tailored to distinct atmospheres (Martínez-García & Hernández-Lemus, 2022).

3) Texture Modality: The 110 texture samples, each enlarged to dimensions of '(224, 224, 3)', provide many tactile characteristics, including "striped" and "polka-dotted." These textures need a CNN to discern the intricate patterns and material characteristics. The uniformity in preprocessing across pictures and textures facilitated easy integration inside the multi-modal framework, enhancing cross-modality feature alignment (Martínez-García & Hernández-Lemus, 2022). The results guided the selection of a cohesive preprocessing strategy and a multi-branch model architecture that caters to the distinct attributes of each modality.

C. Hypotheses

Initial results from the EDA led to the formulation of various hypotheses on the potential enhancement of recommendation quality for home furnishings via the integration of multi-sensory inputs—visual, aural, and tactile.

1) Visual Impact Hypothesis: We propose that the integration of visual attributes (from the picture modality) would markedly enhance suggestion relevancy, since customers often emphasise aesthetics in furniture selection. Comprehensive visual data enables the model to discern colour, shape, and design preferences, which are essential for consumers in choosing things that correspond with their interior aesthetic (Martínez-García & Hernández-Lemus, 2022).

2) Auditory Enhancement Hypothesis: Incorporating ambient sound preferences via audio features would enhance user happiness by harmonising furniture selections with the preferred ambiance. Users who choose a tranquil atmosphere, characterised by noises such as rain or gentle natural sounds, may react well to furniture that complements these settings, hence enhancing the immersive quality of the suggestion (Mateus, 2022).

3) Tactile Influence Hypothesis: Texture information will enhance personalisation by mirroring users' material preferences (e.g., smooth, rough, or patterned surfaces). This tactile data will assist the model in recommending things that align with certain tactile preferences, which are crucial for furniture selection (Mateus, 2022). This study employs a multi-modal deep learning model to amalgamate visual, aural, and tactile input, therefore establishing a holistic recommendation system that encompasses many aspects of user preferences. Conventional recommendation systems typically use a single modality, such as visual data or text, to propose goods. Nonetheless, single-modal systems sometimes inadequately encompass the whole sensory experience, particularly in areas such as home furnishings, where customers evaluate many elements including aesthetics, atmosphere, and tactile quality. A multi-modal model mitigates these constraints by individually processing each sensory input, extracting distinct properties from each, and then integrating these information to provide a comprehensive suggestion (Nagaraja & S, 2023).

VII. Data Analytics

A. Modeling Approach and Rationale for a Multi-Modal Deep Learning Model

The justification for using a multi-modal deep learning model stems from the intricate nature of user preferences, which are seldom influenced by a single cause. For instance, while visual aesthetics are crucial, they may be inadequate if the furniture's texture fails to satisfy the user's tactile preferences or if the item does not correspond with the intended space atmosphere. By integrating picture, audio, and texture data, the model can encapsulate these nuanced preferences, providing suggestions that more accurately align with the user's real-world expectations. This multi-sensory method enables the system to comprehend intricate human requirements, including visual aesthetics, tactile materials, and audio environments, therefore augmenting personalisation and user happiness (Petit et al., 2018).

Each modality (images, audio, and textures) is processed using a specialised sub-network intended to extract pertinent elements. The image sub-network, a convolutional neural network (CNN), emphasises the acquisition of colour, form, and style; the audio sub-network extracts Mel-frequency cepstral coefficients (MFCCs) to analyse ambient sounds; and the texture sub-network, also a CNN, captures patterns and tactile characteristics. The attributes are then integrated via a fusion layer, facilitating cross-modal interactions that allow the model to ascertain the ideal amalgamation of sensory inputs for recommendations. This framework not only offers a more enriched, contextually relevant experience but also corresponds well with users' perceptions and selections of furniture, making the multi-modal deep learning model a formidable option for this application (Petit et al., 2018).

B. Integration of Subnetwork

The architecture of the multi-modal recommendation model has three unique sub-networks: image, audio, and texture, each designed to analyse and extract characteristics from its corresponding modality. The sub-networks operate autonomously to record distinct features of each sensory input, which are then amalgamated via a fusion layer to provide a cohesive representation that informs the final decision (Rawat & Rawat, 2015).

1) Image Sub-Network: The image sub-network processes visual input, collecting critical aesthetic attributes such as colour, form, and design patterns. It employs a convolutional neural network (CNN) architecture, which is particularly effective for picture data because of its capacity to identify spatial hierarchies and patterns. The sub-network begins with an input layer that accepts pictures scaled to '(224, 224, 3)', a standard dimension for image processing applications. Subsequent convolutional layers are implemented, each with progressively larger filter sizes to capture both low-level features (edges, textures) and high-level features (shapes, object components). Every convolutional layer is succeeded by a MaxPooling layer to reduce spatial dimensions, preserving just the most significant features. The convolutional layers' final output is flattened and sent via a dense layer, yielding a concise feature vector that encapsulates the visual characteristics of the furniture item (Shanthakumar, 2020).

2) Audio Sub-Network: The audio sub-network is designed to capture the aural environment of furniture arrangements, processing sounds such as natural ambient noises. Audio data are first converted into Mel-frequency cepstral coefficients (MFCCs), which are extensively used in audio analysis because to their ability to encapsulate spectral characteristics vital for differentiating sound genres. The sub-network starts with an input layer that receives the 13-dimensional MFCC feature vector for every audio sample. The architecture has two thick layers using ReLU activation functions to process and generate embeddings for the audio elements. These layers mitigate the complexity of audio data while maintaining essential auditory attributes, which might affect consumer preferences in choosing furniture with certain ambiance settings (Sreevidya et al., 2024).

3) Texture Sub-Network: The texture sub-network is a CNN-based architecture intended to extract haptic characteristics from photographs depicting various textures. Similar to the image sub-network, the texture sub-network employs convolutional layers to discern patterns and structures characteristic of diverse textures (e.g., striped, smooth, dotted). The system accepts scaled photos of dimensions '(224, 224, 3)', processes them via convolutional and pooling layers, and ultimately flattens the output into a dense representation. The CNN architecture of this sub-network allows for the identification of nuanced patterns that correspond to consumers' material preferences, including smooth or rough textures (Sreevidya et al., 2024).

4) Integration of Sub-Networks through Fusion Layer: After each sub-network generates its feature embedding, these embeddings are integrated in a fusion layer. This integration stage amalgamates the outputs of the image, audio, and texture sub-networks, producing a singular, multi-dimensional feature vector. The vector is then processed via fully connected layers to ascertain the ideal amalgamation of sensory inputs, using a dropout layer to mitigate overfitting. The final layer produces a recommendation score, synthesising visual, aural, and tactile input into a unified prediction that represents user preferences across several sensory dimensions. This multi-modal integration allows the model to provide suggestions that take into account the aesthetic, atmosphere, and material characteristics of furniture (Tacca, 2011).

C. Cross-Layer Attention

Layer (type)	Output Shape	Param #	Connected to
image_input (InputLayer)	(None, 224, 224, 3)	0	–
texture_input (InputLayer)	(None, 224, 224, 3)	0	–
conv2d_8 (Conv2D)	(None, 222, 222, 32)	896	image_input[0][0]
conv2d_10 (Conv2D)	(None, 222, 222, 32)	896	texture_input[0]…
max_pooling2d_8 (MaxPooling2D)	(None, 111, 111, 32)	0	conv2d_8[0][0]
max_pooling2d_10 (MaxPooling2D)	(None, 111, 111, 32)	0	conv2d_10[0][0]
conv2d_9 (Conv2D)	(None, 109, 109, 64)	18,496	max_pooling2d_8[…
conv2d_11 (Conv2D)	(None, 109, 109, 64)	18,496	max_pooling2d_10…
max_pooling2d_9 (MaxPooling2D)	(None, 54, 54, 64)	0	conv2d_9[0][0]
audio_input (InputLayer)	(None, 13)	0	–
max_pooling2d_11 (MaxPooling2D)	(None, 54, 54, 64)	0	conv2d_11[0][0]
flatten_4 (Flatten)	(None, 186624)	0	max_pooling2d_9[…
dense_13 (Dense)	(None, 32)	448	audio_input[0][0]
flatten_5 (Flatten)	(None, 186624)	0	max_pooling2d_11…
dense_12 (Dense)	(None, 64)	11,944,000	flatten_4[0][0]
dense_14 (Dense)	(None, 64)	2,112	dense_13[0][0]
dense_15 (Dense)	(None, 64)	11,944,000	flatten_5[0][0]
concatenate_2 (Concatenate)	(None, 192)	0	dense_12[0][0], dense_14[0][0], dense_15[0][0]
dense_16 (Dense)	(None, 128)	24,704	concatenate_2[0]…
dropout_2 (Dropout)	(None, 128)	0	dense_16[0][0]
dense_17 (Dense)	(None, 64)	8,256	dropout_2[0][0]
output (Dense)	(None, 1)	65	dense_17[0][0]

Fig. 4. multimodal model architecture using Keras

This multi-modal recommendation model's design includes essential technological decisions to enhance its capacity to process and integrate various sensory inputs, such as cross-layer attention and specialised neural network topologies for each modality (Vaswani et al., 2017). A significant concern was the prospective use of cross-layer attention mechanisms, a method enabling the model to choose concentrate on critical elements across several layers and modalities. Cross-layer attention dynamically allocates weights to features according to their significance, allowing the model to emphasise certain elements of the input during integration. For example, when suggesting furniture that complements a cosy atmosphere, the audio attributes reflecting ambient noises (such as rain or quiet

music) may be emphasised, but visual attributes may take precedence for suggestions focused on aesthetics (Vaswani et al., 2017). This method augments the model's flexibility, enabling it to prioritise certain sensory inputs according to the user's implicit preferences, hence enhancing the personalisation and relevancy of the suggestions.

1) Neural Network Configurations: The neural network architecture of the model is meticulously tailored for each modality, guaranteeing efficient processing of each sensory input.

Convolutional Neural Networks (CNNs) are used in the image and texture sub-networks. Convolutional Neural Networks (CNNs) are proficient at identifying spatial patterns and hierarchies, crucial for discerning the forms, colours, and textures of furniture. Pooling layers were included subsequent to each convolutional layer to decrease dimensionality while preserving the most salient features (Vats et al., 2024).

The dense layers in the audio sub-network process the Mel-frequency cepstral coefficients (MFCCs) derived from audio. Dense layers were used in this instance because the simplicity of MFCC data allows a basic fully connected network to successfully capture the essential spectrum information without the intricacies of convolutions (Wu et al., 2022).

2) Regularization Techniques: Dropout layers were included subsequent to the fusion layer to mitigate overfitting. This regularisation method randomly excludes certain characteristics during training, enhancing the model's ability to generalise to novel data. These selections together guarantee that the model can efficiently handle each modality while dynamically adjusting to changing user preferences, therefore establishing a robust and adaptable multi-modal recommendation system (Wu et al., 2022).

VIII. Data Visualization and Results Report

A. Training and Evaluation results

The training procedure for the multi-modal recommendation model used a compact, balanced dataset divided into training and validation subsets. The model underwent training for 10 epochs with a batch size of 8, facilitating the processing of data in small groups to enhance computing efficiency and gradient updates. Each epoch signifies a complete traversal of the training dataset, during which the model tries to discern patterns from each modality: pictures, audio, and textures (F. Zhang et al., 2023). The training set for each modality included 35 samples, providing the model with a basic dataset to understand cross-modal correlations and sensory preferences. The validation set included 9 samples for each modality: pictures, audio, and textures. The validation set was used to assess the model's performance on novel data, offering insight into its generalisation capabilities.

Training set:	**35 images, 35 audio samples, 35 textures**
Validation set:	**9 images, 9 audio samples, 9 textures**

TABLE II
Data Integration and Splitting

B. Accuracy and Loss Plots

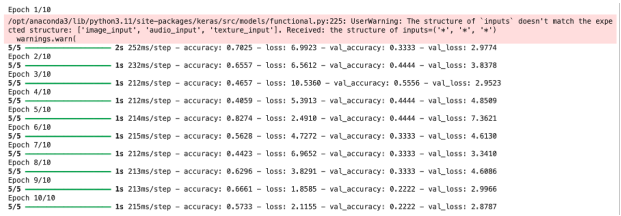

Fig. 5. Training the Model

The accuracy and loss graphs for both training and validation datasets elucidate the model's learning trajectory throughout 10 epochs. The training accuracy exhibited considerable fluctuation, spanning from about 40% to 80%, but the validation accuracy ranged from 30% to 55%. The oscillations suggest difficulties with generalisation, perhaps resulting from the restricted sample size. Limited datasets may result in significant fluctuation in model performance, since the model may lack sufficient different instances to establish stable, generalisable patterns (F. Zhang et al., 2023). The training

Fig. 6. Training and Validation Accuracy over Epochs (Left), Training and Validation Loss over Epochs (Right)

loss exhibits a consistent decline, indicating that the model is effectively learning to reduce mistakes in the training data. Nevertheless, the validation loss exhibited considerable oscillation, lacking a continuous decreasing trajectory, suggesting either overfitting or susceptibility to slight differences in the validation samples. This phenomenon is prevalent in models trained on limited datasets, where the model may excel on the training set but falter in generalising to novel data (S. Zhang et al., 2019). The downward training loss indicates the model's proficiency in identifying connections within the training data. The fluctuations in validation measures highlight the need for a more extensive dataset to get more consistency and stability in generalisation. These results suggest that while the model has learning potential, its dependability in practical applications would be greatly enhanced with a more comprehensive and varied dataset (S. Zhang et al., 2019). The training and validation curves for this multi-modal recommendation model provide an in-depth analysis of the model's learning progression across 10 epochs, highlighting its strengths and limits in generalising to novel data. By analysing

these patterns, we can evaluate the model's conformity with original assumptions and pinpoint areas for improvement.

C. Interpretation of results and findings

The training accuracy and loss curves indicate that the model successfully learnt patterns from the training data across each epoch. The training accuracy typically varied between 40% and 80%, exhibiting oscillations among epochs. The training loss had a declining trajectory, indicating that the model was reducing mistakes and adjusting to the configuration of the training data. This pattern is anticipated in deep learning models, since they seek to minimise the loss function and optimise accuracy over time, especially during the first epochs. These data correspond with the expected learning curve, whereby training loss generally decreases as the model acquires knowledge (S. Zhang et al., 2019).

Nonetheless, the variations in training accuracy indicate that the model did not exhibit consistent improvement across epochs. The discrepancies are probably due to the limited dataset, which fails to provide sufficient sample diversity for the model to attain stable learning. When data is scarce, the model may exhibit sensitivity to certain samples, resulting in performance fluctuation when faced with distinct data subsets. This pattern illustrates a significant constraint in training on a limited dataset, whereby model dependability is undermined by the narrow spectrum of samples.

1) Interpretation of Validation Trends: The validation accuracy and loss curves provide more insights into the model's generalisation abilities. The validation accuracy varied between 30% and 55%, whereas the validation loss exhibited oscillations without a discernible decreasing trajectory. The variations and irregularities in the validation measures indicate that the model had difficulties in generalising to new, unknown data, a prevalent issue when training on restricted samples. In contrast to the training loss, which exhibited a steady decline, the validation loss had significant volatility, indicating that the model's performance on the validation set was inconsistent (S. Zhang et al., 2019).

The significant fluctuation in validation loss indicates probable overfitting. Overfitting transpires when a model assimilates the intricacies and noise of the training data to such a degree that it adversely affects performance on unseen data. Although the model identified links among the training data, it may have concentrated excessively on specific patterns, lacking a more comprehensive, generalised knowledge. The dropout layers used during training sought to alleviate this danger; nevertheless, because to the sparse data, the regularisation impact of dropout alone was unable to entirely avert overfitting (S. Zhang et al., 2019).

2) Comparison to Initial Expectations: The model was anticipated to demonstrate consistent improvement in both training and validation performance, attaining reasonably stable metrics by the concluding epochs. Nonetheless, the findings indicate that, while the model acquired knowledge from the training data, it failed to generalise proficiently. The original assumption that the model could achieve consistent generalisation with a limited sample set was too optimistic. Multi-modal models, particularly deep neural networks, often get advantages from large datasets owing to the intricate nature of pattern recognition across varied inputs (images, sounds, and textures). The restricted data quantity hindered the model's capacity to acquire strong cross-modal associations, which are crucial for a comprehensive recommendation system (Zhao et al., 2024).

3) Summary and Recommendations: The model's results underscored the difficulties of training a multi-modal model with little data. While the training curves indicated learning, the validation curves highlighted generalisation problems. The discrepancy between training and validation performance indicates that obtaining consistent and dependable suggestions requires a bigger and more varied dataset. Augmenting the data would provide the model with more instances, so improving its capacity to identify wider patterns and diminishing its susceptibility to particular samples. Moreover, the integration of additional regularisation methods, such as early halting or data augmentation, may enhance generalisation. In conclusion, while the model's preliminary learning was encouraging, the patterns suggest that further data and enhancements are necessary for it to achieve its full potential in practical applications (Zhao et al., 2024).

IX. Conclusion

To improve home furnishings recommendations, this research developed a multi-modal recommendation system including visuals, music, and textures. This recommendation system uses various sensory data types to match users' visual, aural, and tactile preferences, unlike single-modal systems. The study examined the pros and cons of multi-sensory recommendation systems using a deep learning model.

Experimental findings from multi-modal model training showed promise and limits. Unique discoveries came from each method. The picture sub-network recorded colours, forms, and design features, which people evaluate while choosing home furniture. Mel-frequency cepstral coefficients (MFCCs) represented noises in the audio sub-network to satisfy consumers who want certain atmospheric characteristics in their homes. The texture sub-network helped the model recognise tactile cues and match consumers' furniture material and finish preferences.

Training loss decreased consistently, indicating that the model could capture correlations in training data. Due to the little dataset, validation accuracy and loss oscillated, showing the model struggled with generalisation. This shows that deep learning models trained on short datasets may overfit to the training data without learning generally applicable features.

The results suggest that multi-modal recommendation systems may improve decision-making, especially in sensory domains. In a real shop, consumers see, feel, and hear the environment surrounding the furniture. The model simulates this by merging visual, aural, and tactile data. This upgraded method offers a more immersive and personalised experience

than visual-based suggestions. Multi-sensory recommendation systems can meet complicated customer preferences and boost user happiness and engagement for companies and e-commerce platforms. Visual and tactile qualities of places and materials may impact consumers in other industries like tourism and fashion.

This work integrates many sensory modalities into a recommendation system model. Multi-modal recommendation systems are developing, although most use textual and visual data. This study extends multi-sensory recommendation systems by adding auditory and tactile data and emphasises the importance of combining diverse sensory information to capture complicated user preferences. Convolutional neural networks (CNNs) and dense layers may be used for multi-modal applications using sub-networks for each modality, according to the research. This modular design lets domain-specific sensory inputs be added or modified. Multi-modal models are computationally costly yet provide a more accurate knowledge of user preferences for real-world decision-making.

The research had promise, but various restrictions limited the model's effectiveness and generalisability. The dataset was limited, with 35 training examples and 9 validation samples per modality. This restricted data amount caused overfitting, where the model learnt precise features from training data without catching broader trends. The validation performance was uneven, making it hard to evaluate the model's potential. The research used artificially produced data divides and did not include user preferences or input, which are necessary to validate suggestion quality in practice. A model trained and assessed on real user data may provide more accurate insights into its efficacy. This research focused on visuals, music, and textures, but product descriptions, user ratings, and fragrance data for perfumes and cosmetics might improve the system. Extending sensory inputs would demand a more complicated model architecture and bigger dataset. Multi-modal models demand a lot of processing resources, especially when training on high-dimensional input like pictures and music. Limited resources may limit hyperparameter tweaking, model setups, and sophisticated approaches like cross-modal attention.

Given these limitations, various future research initiatives might reinforce and broaden this study's conclusions. Increasing the dataset size improves model generalisability and performance consistency. More instances and a bigger, more varied dataset would help the model discover generally applicable patterns and reduce overfitting. Real-world user data, like as preferences or interaction logs, might also verify the model in practice. Another essential method for model refining is user input. input loops enhance real-world recommendation systems by using user input to modify suggestions. Future research might employ reinforcement learning to enhance suggestion relevance by learning from user behaviour via a feedback system.

Exploring more sensory modalities might improve the system. This study employed three sensory modalities, but future research might use more. For instance, adding textual data like product descriptions or user reviews or sensory clues like aroma or taste for certain domains might help identify consumer preferences. Adding these modalities would require a more sophisticated model design but might capture more nuanced user preferences. Advanced fusion research is also promising. Future research might use cross-layer attention to dynamically concentrate on key characteristics across modalities. The model might emphasise user-intented sensory inputs via cross-layer attention, improving suggestion relevance. Transformers can capture long-range relationships between sensory components, which may improve multi-modal fusion. Additionally, computing efficiency must be improved. Due to their high computational needs, multi-modal models must be optimised for efficiency. Model pruning, quantisation, and transfer learning might make multi-modal models more practical and scalable without losing performance. Finally, this research lays the groundwork for incorporating multi-sensory data into recommendation systems and reveals the promise of multi-modal deep learning models for personalised suggestions. While generalising from little data is difficult, the technique shows potential for enhancing user experiences in sensory-intensive fields. Expanding this study to overcome dataset, architectural, and computational problems might enable recommendation systems to provide more personalised and engaging user experiences. article [utf8]inputenc hyperref

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